

SDIE V6 — Combined ML Metrics Report

Train corpus (X): **Tagged Files_2** | Test corpus (Y): **Raw files_2** | Generated: 2026-06-10

Phase target: classification F1 and reference accuracy $\geq$ **85%**

Combined ML Metrics — Overview

	Train (X)	Test (Y)	Target
Corpus	Tagged Files_2	Raw files_2	—
Drawings	47	18 (18 paired)	—
Metric type	Entity-level GT (manifest flags)	Paired teach-reference proxy	—

Sample Counts

	Train (X)	Test (Y)
Ground truth / reference	25,761 slab+beam entities	Teach ref: 9,641 slab + 10,734 beam
Total classified (with prediction)	22,943	14,625
Missing predictions	2,818	—
Predicted as Slab	11,438 (5,269 correct TP)	1,887 labeled / 1,709 ref-count
Predicted as Beam	9,017 (3,609 correct TP)	10,687 labeled / 9,273 ref-count

Correctness

	Train (X)	Test (Y)
Correct (any class)	8,878	N/A (no entity-level GT on raw)
Correct Slab	5,269 / 11,767 GT	Mean ref entity-count acc 27.2%
Correct Beam	3,609 / 13,994 GT	Mean ref entity-count acc 82.4%

Accuracy & F1

Metric	Train (X)	Test (Y)	Target
Overall accuracy	38.7%	37.0%	$\geq$ 85%
Slab F1 / ref acc	45.4%	27.2%	$\geq$ 85%
Beam F1 / ref acc	35.7%	82.4%	$\geq$ 85%
Slab+Beam mean	40.5%	—	$\geq$ 85%
Pass	No	0/18 files	—

Train (X) — Detail

- **47** teach drawings (25,761 slab+beam GT entities)
- **22,943** entities received a prediction; **2,818** had none

- **8,878** correct overall (5,269 slab TP + 3,609 beam TP)
- Primary error mode: slab↔beam confusion (5,408 slab GT → predicted beam; 6,169 beam GT → predicted slab)
- Drawings 1–10: cached DeepSeek run; drawings 11–47: --no-deepseek (API timeouts)
- Quality gate: **FAIL** (accuracy 38.7%, mean F1 40.5%)

Test (Y) — Detail

- **18/18** raw DXFs paired to teach references
- **14,625** entities classified total
- **1,887** predicted Slab, **10,687** predicted Beam (classification proxy)
- vs teach reference counts: slab 1,709 actual vs 9,641 expected; beam 9,273 vs 10,734
- Beam count tracking relatively strong (~82.4%); slab under-detected (~27.2% mean ref acc)
- **0/18** files passed the 85% quality gate

Grand Totals (Both Splits)

	Count
Entities classified (train + test)	37,568
Correct Slab (train TP only)	5,269
Correct Beam (train TP only)	3,609
Correct Slab+Beam (train only)	8,878

Notes: Test metrics use paired teach-reference count accuracy, not per-entity TP/FN like train. Test data source: *git archive (431aa01:P2_SlabVersion6/Output/ml_eval/test_metrics.json)*. Re-run `python scripts/evaluate_ml_project.py --test-only` to refresh test metrics on disk.

Source files: `Output\ml_eval\train_metrics.json` | `test_metrics` (archived)